Lab experiment-3

Water Jug Problem

Aim

To write a Python program to solve the Water Jug Problem.

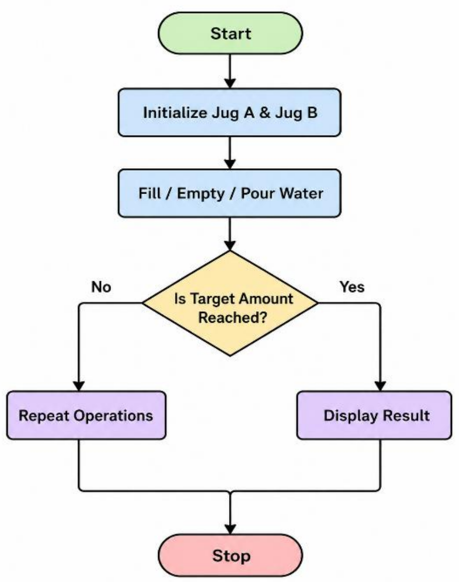
Objective

To find a sequence of steps to obtain the required amount of water using two jugs of different capacities.

Algorithm

1. Start.
2. Initialize two jugs with capacities (e.g., 4 L and 3 L).
3. Fill, empty, or pour water between the jugs.
4. Repeat the operations until the target amount is obtained.
5. Display each step of the solution.
6. Stop.

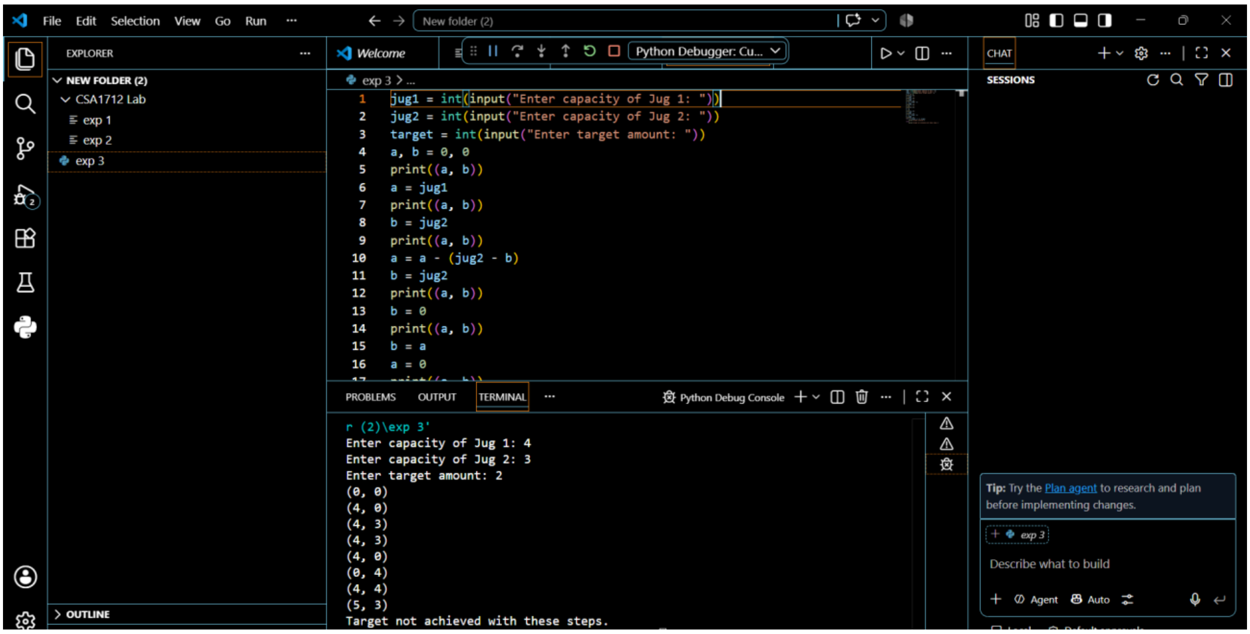
Flow chart



Code

```
jug1 = int(input("Enter capacity of Jug 1: "))
jug2 = int(input("Enter capacity of Jug 2: "))
target = int(input("Enter target amount: "))
a, b = 0, 0
print((a, b))
a = jug1
print((a, b))
b = jug2
print((a, b))
a = a - (jug2 - b)
b = jug2
print((a, b))
b = 0
print((a, b))
b = a
a = 0
print((a, b))
a = jug1
print((a, b))
a = a - (jug2 - b)
b = jug2
print((a, b))
if a == target or b == target:
    print("Target achieved!")
else:
    print("Target not achieved with these steps.")
```

Output



Result

The program executed successfully and obtained the required 2 liters of water.

Conclusion

The Water Jug Problem was solved successfully using Python. The desired amount of water was obtained using valid jug operations.